SmartStep Use Case Diagram and User Interface

The SmartStep User Interface (UI) is designed to be an engaging and accessible platform for students and staff at CTU Argao. Its primary goal is to promote "green" participation by visualizing how simple daily activities—like walking to class—contribute to the sustainable energy goals.

General

Fig 1.1 Whole Use Case Diagram

This is the complete use case diagram showing the interactions of two actors — Admin and User — with the entire system. Both actors authenticate via log in/log out, after which they access their respective features. The Admin manages system-level functions (dashboard analytics, advanced analytics, system configuration), while the Actor accesses user-facing features (interactive dashboard, general analytics, user settings).

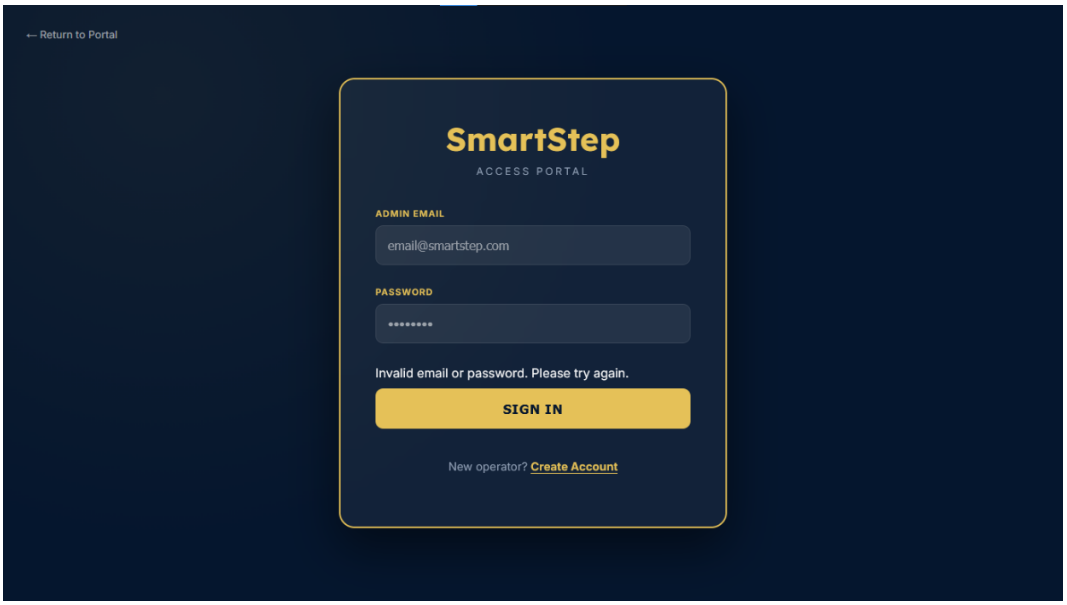

Fig 1.2 Log In Interface

The Log In screen is the primary access point for both Administrators and General Users. It is designed with a clean, high-contrast interface that requires a registered Username and Password to enter the dashboard. This authentication layer is critical for our project because it ensures that sensitive hardware controls—like battery thresholds and system restarts—are protected from unauthorized access, maintaining the overall security of the IoT network.

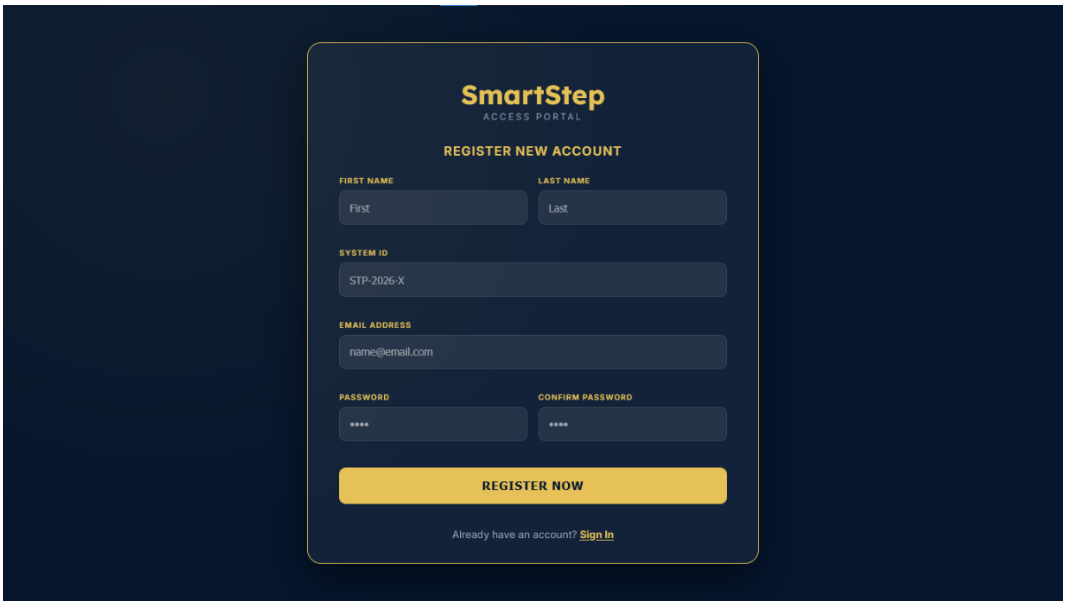

Fig 1.3 Register Interface

The Register screen allows new members of the CTU faculty or the development team to create an account and join the SmartStep ecosystem. It captures essential user details, including Full Name, Email, and a unique Username. To ensure that only legitimate users can sign up, the registration process is tied to our backend database, allowing us to manage user roles and track who is interacting with the system's data and hardware settings.

User

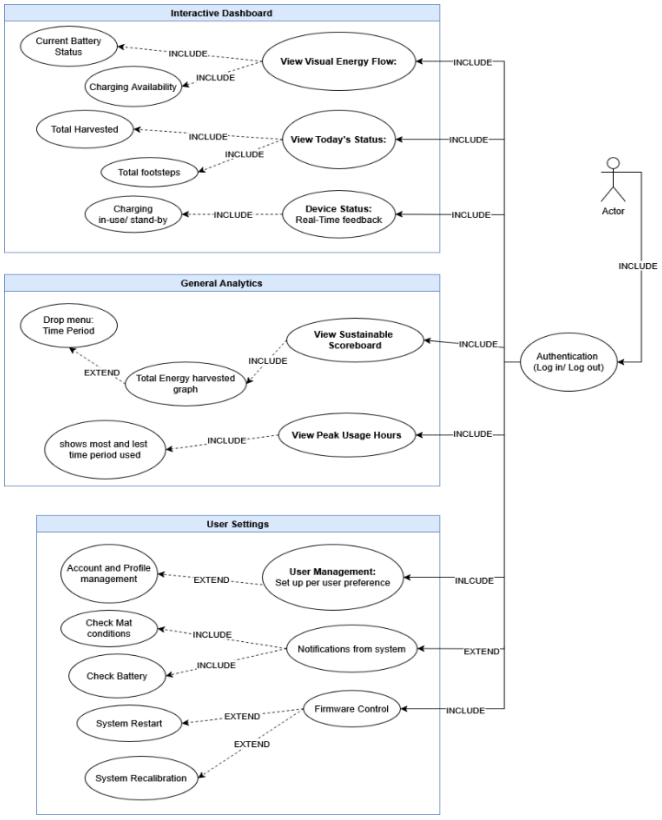

Fig 2.1 User Use Case Diagram

User Use Case Diagram This diagram focuses on the User (regular user) and their interactions after authentication. The user can access three modules: Interactive Dashboard (visual energy flow, today's status, device real-time feedback), General Analytics (sustainable scoreboard, peak usage hours, energy harvest graphs), and User Settings (profile management, system notifications, firmware control with restart/recalibration options).

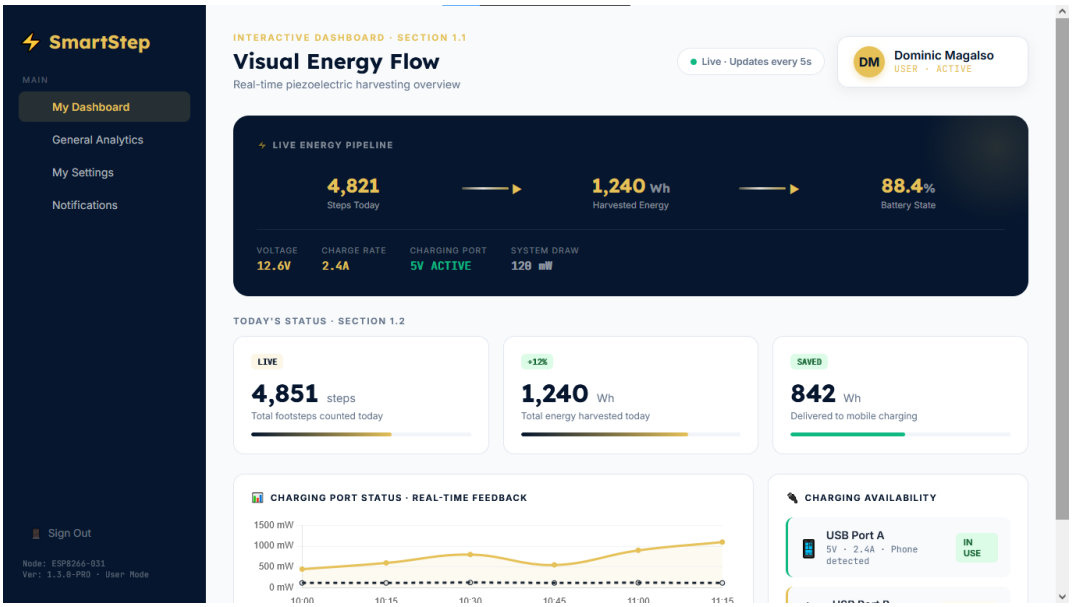

Fig 2.2 User Dashboard

The walkthrough begins with the User Dashboard, the primary screen for everyday interaction. Unlike the technical admin view, this screen emphasizes availability and personal impact. At a glance, the user can see the Visualized Energy Flow, which shows the current battery status of the charging station. The Integrated Dashboard serves as the main hub for live system tracking. At its center, the Energy Flow animation confirms that the PZT mat is actively harvesting power and sending it to the battery. Precision gauges for State of Charge (SoC) and Wattage provide an instant "health check" of the system, ensuring the storage cells have enough energy to support student charging needs.

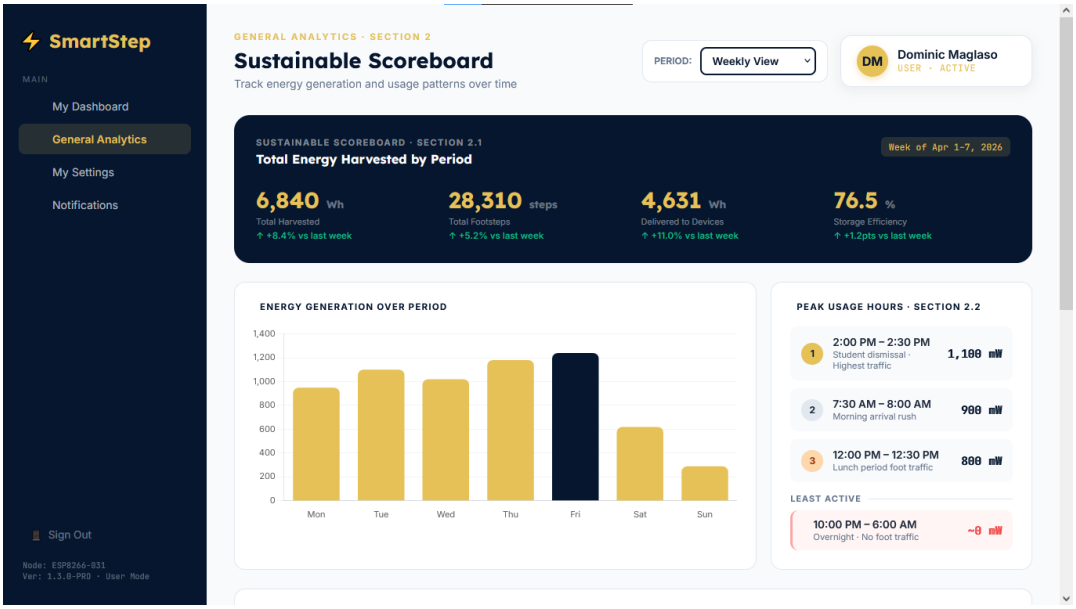

Fig 1.3 User Analytics

Moving to the Analytics section, the user is presented with their Personal Footprint. This module tracks "Today's Steps," giving the user immediate feedback on how their movement has contributed to the system's reservoir. To foster a sense of community, the Sustainability Scoreboard displays the total energy harvested by the entire campus over the week or month. By showing these cumulative milestones, the GUI transforms a technical hardware project into a shared environmental mission, encouraging more people to use the SmartStep mats.

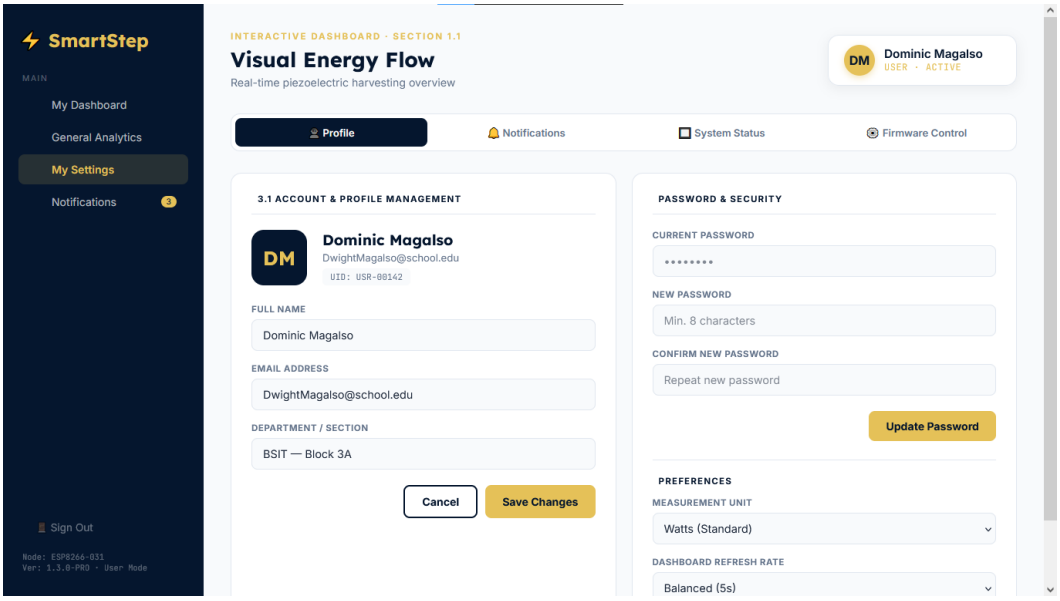

Fig 1.4 User Account Settings

The walkthrough concludes with the User Settings module. Here, students and staff can manage their personal profiles and customize their notification preferences. This section allows users to opt-in for "Green Milestones" alerts or adjust how their data is displayed on the community scoreboard. This level of personalization ensures that the SmartStep system remains a user-centric tool that rewards sustainable behavior while respecting individual user privacy and preferences.

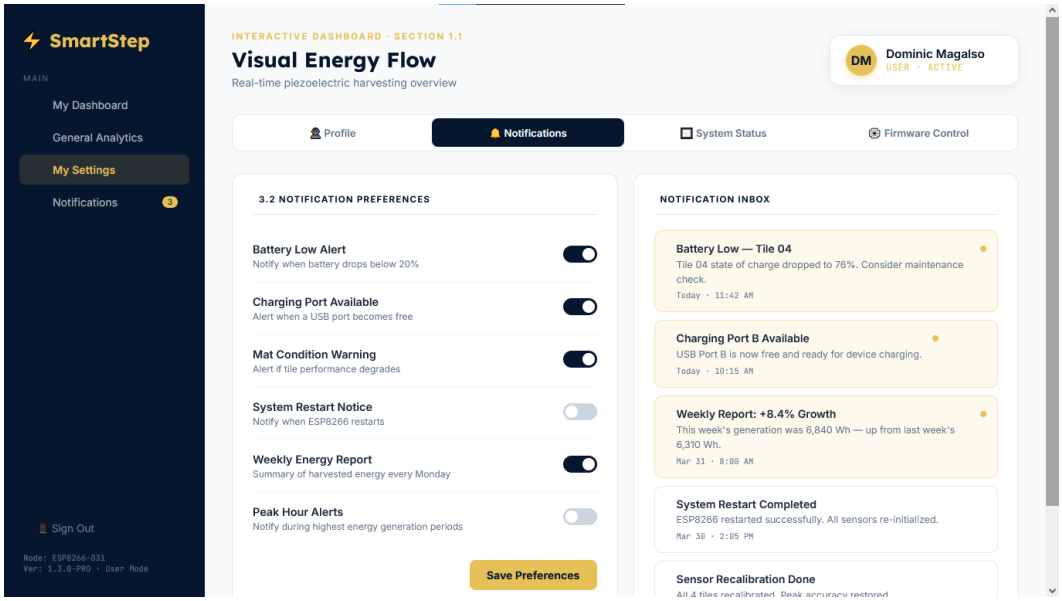

Fig 1.5 User Notifications

The Notifications hub on the user side is streamlined for utility rather than technical diagnostics. Users receive helpful alerts such as "Station Available," "Battery Low," or "Maintenance in Progress." These notifications ensure that the student body is always informed about the station's status. For example, if the battery is depleted after a period of high demand, a "Battery Low" alert managed by the system informs the user why charging might be temporarily unavailable, managing expectations effectively.

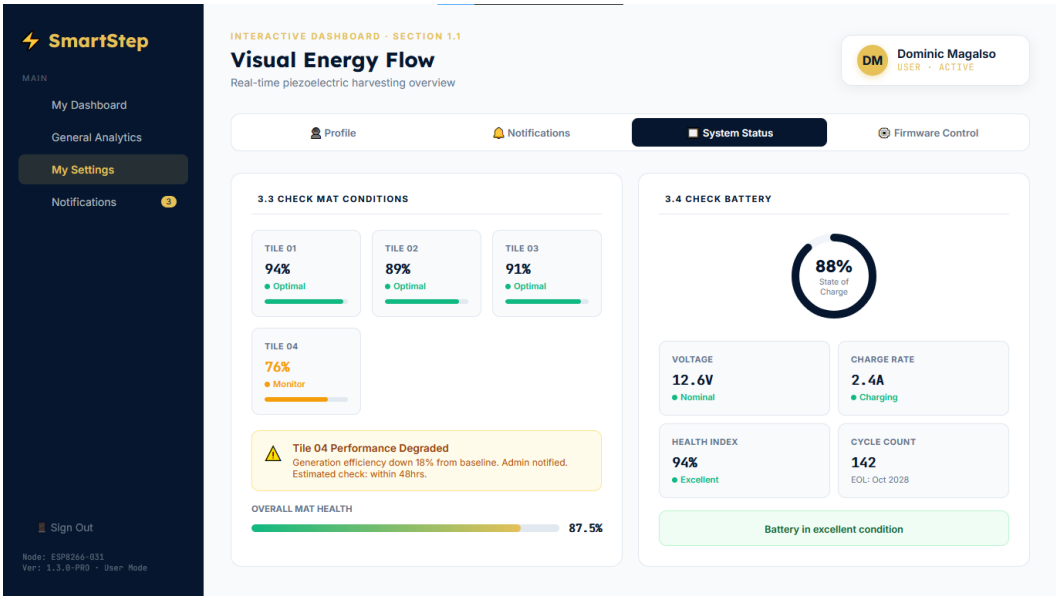

Fig 1.6 User System Status

The Hardware Status section acts as the system's diagnostic tool. It monitors the physical integrity of the PZT sensor array through a Mat Condition gauge, ensuring any mechanical wear is caught early. Additionally, it tracks the ESP8266 Signal Strength (RSSI) and Internal Temperature, verifying that the controller remains connected and operating within safe thermal limits.

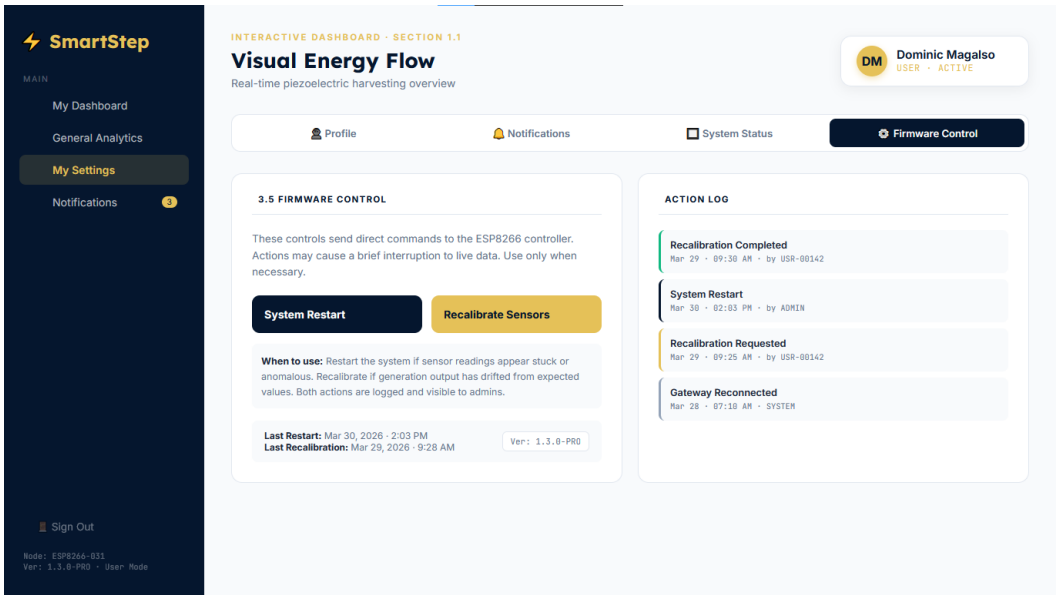

Fig 1.7 User Remote Firmware Config

The Remote Config panel allows for direct management of the system's logic. Through this interface, we can trigger a System Restart or Sensor Recalibration to maintain data accuracy. Most importantly, it features Voltage Threshold Settings, which act as a digital safety net. By setting these cut-off points, the firmware automatically protects the battery from deep discharge, ensuring the hardware lasts longer.

Admin

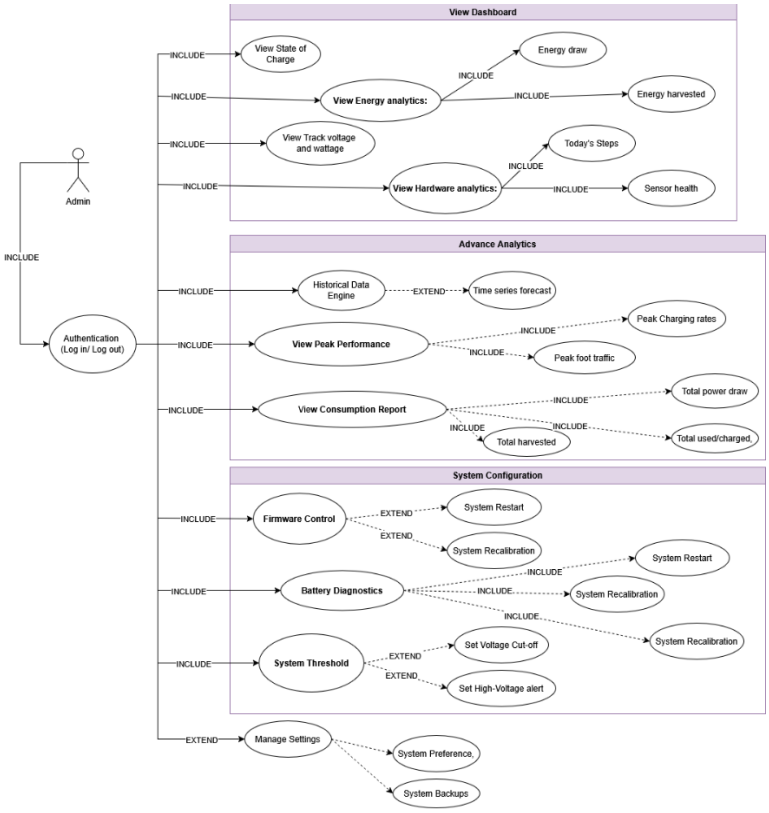

Fig 2. Admin Use Case Diagram

This diagram focuses solely on the Admin actor and their interactions with the system. After authentication, the Admin can access three main modules: View Dashboard (energy and hardware analytics), Advance Analytics (peak performance, consumption reports, historical data), and System Configuration (firmware control, battery diagnostics, system thresholds). The Admin can also extend to manage system settings like preferences and backups.

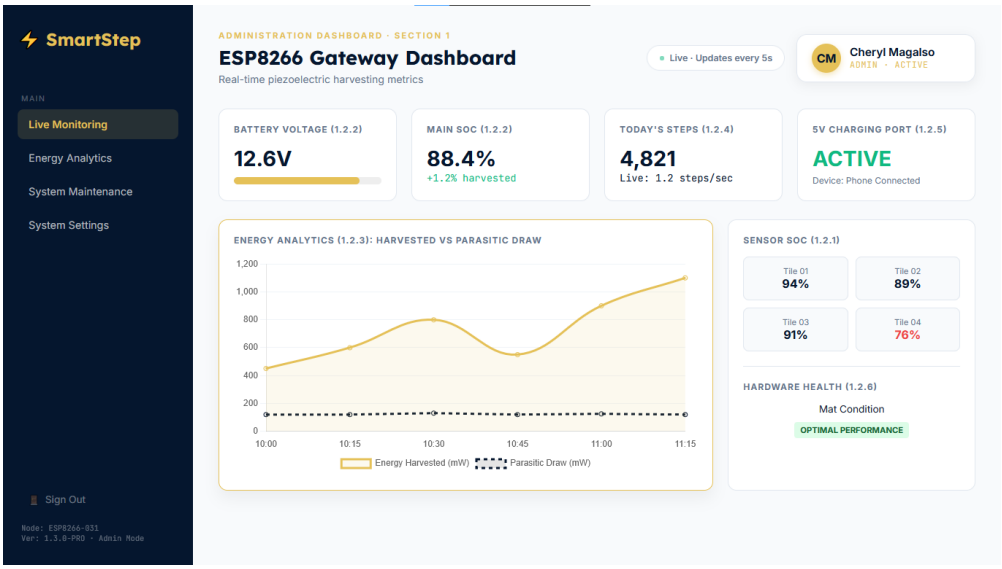

Fig 2.1 Admin Dashboard

The Admin Dashboard provides an immediate high-level overview of the system's vitals. It features live visualizations of Energy Flow from the PZT mat to the storage unit, alongside precision gauges for the State of Charge (SoC) and Real-time Wattage. This page ensures we can monitor current power generation and battery health at a single glance.

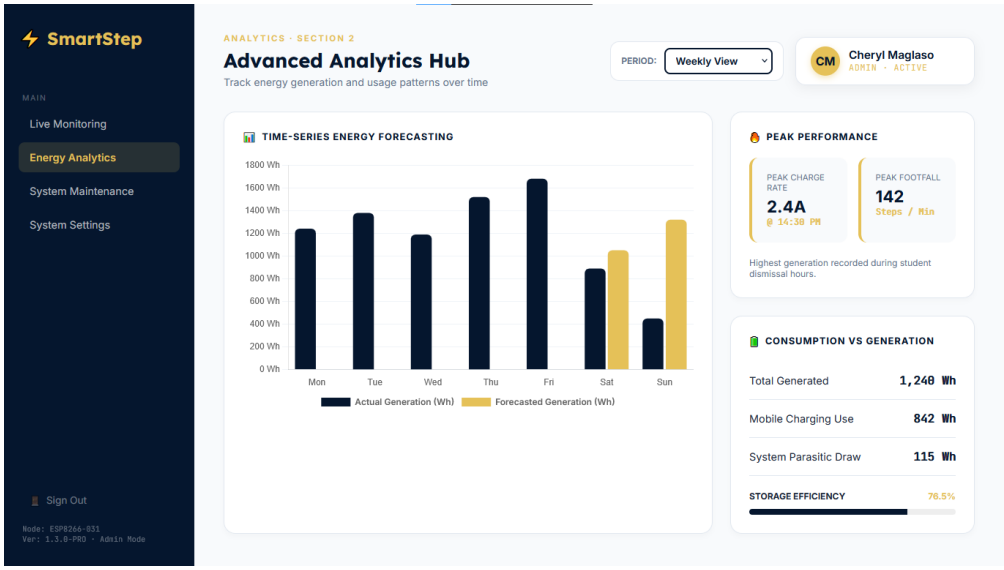

Fig 2.2 Admin Analytics

The Analytics Page shifts focus from "what is happening now" to long-term performance trends. It utilizes detailed line graphs to track Energy Harvested vs. Parasitic Loss and Footfall Frequency. This allows us to evaluate the efficiency of our power conversion and identify the busiest times of day for energy generation on campus.

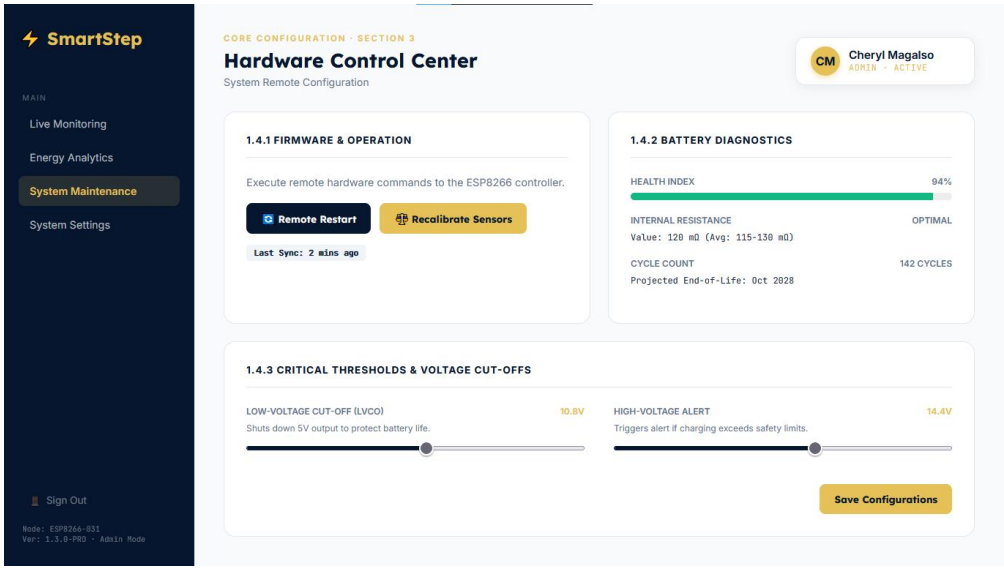

Fig 2.3 Admin Hardware Control

The System Config Page is the technical control center where we manage the "brain" of the SmartStep mat. From here, we can trigger System Restarts or Sensor Recalibrations to keep the hardware accurate. Most importantly, it allows us to set Voltage Thresholds, ensuring the firmware automatically protects the battery from damage by cutting off power if levels drop too low.

The System Status Page provides "under-the-hood" diagnostics of the physical hardware. It tracks the Mat Condition to spot mechanical wear and monitors the ESP8266 Signal Strength (RSSI) and Internal Temperature. This ensures the IoT connection remains stable and the hardware stays within safe thermal operating limit

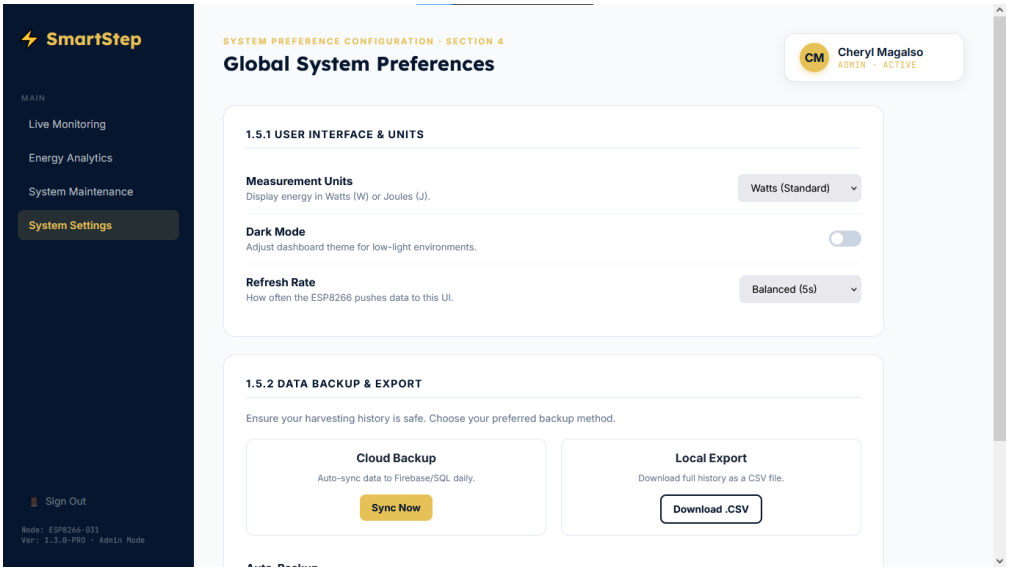

Fig 2.4 Admin System Settings

The Settings Pages manage the human side of the system. These pages allow us to update Admin Profiles, manage security credentials, and configure system-wide preferences. This ensures that faculty can access the sensitive remote configuration and maintenance tools.